

CL27-16k remains the base: CL33 raises Skiing ID by deleting some eyewear, while no CL30–CL37 arm improves aggregate identity

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Scope: completed CL30–CL37 scientific runs; immutable Comet histories and table-sealed validation panels; paired per-image identity analysis; hard-case visual review; loss-activity and configuration audit; comparison with CL27-16k and controlled PhotoMaker. Failed retries and one-item smokes are operational evidence only and are excluded from scientific comparisons.

Primary metric: mask-owned subject-v2 manual_val/id_sim on the fixed 96-image manual_val panel.

Reproducible assets: [assets/cl30_cl37_20260819/](#)

Executive conclusion

Keep CL27 at 16k as the primary architecture and checkpoint for further experiments. Do not promote a complete CL30–CL37 configuration. The closest new arm, CL33 at 16k, reaches 0.546311 ID_SIM versus CL27’s 0.547260. Its paired difference is -0.000949 , 47/96 wins, with a 95% cell-bootstrap interval of $[-0.007764, +0.006280]$: aggregate equivalence is plausible, but improvement is not established. Every other arm is lower, and CL30, CL34, CL36, and CL37 are clearly negative against their correct CL27 controls. [measured] [paired]

CL33 produces the only important hard-case gain: Skiing rises from 0.433680 to 0.461983, almost matching PhotoMaker’s 0.464005. Visual inspection changes the interpretation. Elon gains ID partly because CL33 **deletes his identity-defining ordinary glasses**; Marion’s large goggles still cover the eye region. PhotoMaker remains 8 pass / 0 minor / 0 fail on the eight Skiing cells, compared with 6/1/1 for CL27-16k and 6/0/2 for CL33-16k under the predeclared topology rubric. CL33 improves some goggle layers, but it has learned an object-deletion shortcut and is not a clean occlusion solution. [measured] [visual]

Crying is no longer the limiting topology case: all reviewed selected panels keep hands and faces readable (8/8 passes for PhotoMaker, CL27, and CL33). The remaining weaknesses are the Skiing object/face layer order, Marion ID, and inconsistent small/action faces. None of the new arms improves Marion’s 12-prompt mean over CL27 (0.493482); CL31 is the closest at 0.488494. CL32 has the best new Jumping mean (0.410518), and CL35 the best Dancing mean (0.461924), but neither improvement generalizes to aggregate ID. [measured] [visual]

The mechanism audit prevents two false conclusions. CL35’s DINOv2 patch-ID reward is nonzero only around step 100 and stays exactly zero after 1k because the observed gate mass averages about 0.111 while the configured eligibility floor is 0.55. CL36’s ArcFace hinge is similarly almost absent: identity loss is nonzero in only 4/60 post-1k logged windows and its BA auxiliary gradient ratio is zero throughout. The complete CL35 and CL36 recipes are negative, but these runs do **not** establish that patch identity or ArcFace supervision is intrinsically ineffective. [measured] [code]

PhotoMaker still leads the selected CL27 checkpoint by 0.009320 mean ID. The paired interval $[-0.021757, +0.003438]$ crosses zero on this 96-cell panel, but the point estimate and visual Skiing result both favor PhotoMaker. The next work should therefore preserve CL27’s active frequency-surface BA mechanism, pin the 16k checkpoint as the source/control, and repair one objective at a time rather than stacking any CL30–CL37 recipe. [measured] [decision]

1. Evidence integrity

1.1 Immutable scientific identities and completion

Each scientific row below is the final successful retry. MLS recorded the job as completed with exit code zero, and each run has a table-sealed fixed-96 Comet panel. Earlier failed retries and bounded smokes are not joined by display name and are not used in the analysis. [measured]

Arm	Critical change versus CL27	Immutable Comet key	Selected step	Final step
CL30 r4	positive-only low-band	db38cfb250d241cf89bf57705ff86b18	16k	24k
CL31 r4	same-ID pull attention ownership alignment	ed5077fd3cfc41bd898c1234b8c3ba24	24k	24k
CL32 r1	contact- frequency surface partition	078cf231674f4fa499e160a435300511	18k	24k
CL33 r1	visibility- balanced reconstruction	3173f3086fa344f7ad3eb6ce7b07ac1f	16k	24k
CL34 r4	learned shared frequency schedule	577cc412ffa04e5686e5c10760186c65	18k	24k
CL35 r7	attention-gated DINOv2 patch identity	f3417ee9a86342cb9bc13e5eb37bb3e2	24k	24k
CL36 r4	ArcFace hinge continuation from CL27-16k	41dcb0987d5d439bb14329052953ff6d	4k	4k
CL37 r4	small-face ROI teacher distillation	f3c535315da242d78494d7df6dd1eaa3	18k	24k

Controls are CL27 r3, key `dbfbf40c3bdd4f70bedc58bda3dfb9cd`, selected at 16k, and PhotoMaker step 0, key `74efd227d3f8488a98e83d815c77c07c`. All eligible arms retain branched self-attention, target-query/reference-KV conditioning, `pose_adapt_ratio=0`, and `ca_mixing_for_face=false`. None improves by switching off the BA core. [\[code\]](#)

1.2 Fixed panel, joins, and uncertainty

The analysis uses the fixed 96-image `manual_val` contract: one image per identity/prompt cell with unchanged references, prompts, seeds, ownership boxes, scheduler, inference steps, CFG, and subject-v2 metric. Each selected export has exactly 96 images and one 96-row per-image table with no exporter warning or error. Filenames were normalized for spaces and underscores before joining image, table, and bounding-box records. The table is the completion seal, so a partially uploaded next panel cannot be mistaken for a checkpoint. [\[measured\]](#)

Paired intervals use 50,000 bootstrap resamples of the fixed cells. They quantify uncertainty across this panel, not across training seeds. The visual rubric is separate from ID_SIM:

- **pass:** prompted top object and identity-defining ordinary eyewear are present, object/face order is readable, and the intended face is attached;
- **minor:** readable ownership and order with a localized asymmetry;
- **fail:** fused or duplicated layers, important object deletion, unreadable face, or wrong face/body association.

Visual counts are one unblinded review of one fixed seed. They diagnose cells; they are not a population estimate. [\[limitation\]](#)

1.3 Step-zero parity

CL30, CL31, CL32, CL33, CL35, and CL37 have per-image ID vectors exactly equal to CL27 at step zero (`max |delta| = 0`). CL36 starts from CL27-16k and its step-zero ID vector is exactly equal to that source checkpoint. These results exclude a favorable initialization as the explanation for their later scores. [\[measured\]](#)

CL34 is the exception: step-zero mean is 0.464134 versus CL27’s 0.464640, and the maximum per-cell difference is 0.032371. The change is small relative to CL34’s later deficit, but strict step-zero parity is false; no difference-in-differences causal claim is made for CL34. [measured] [caveat]

2. Quantitative results

2.1 Identity trajectories and paired decisions

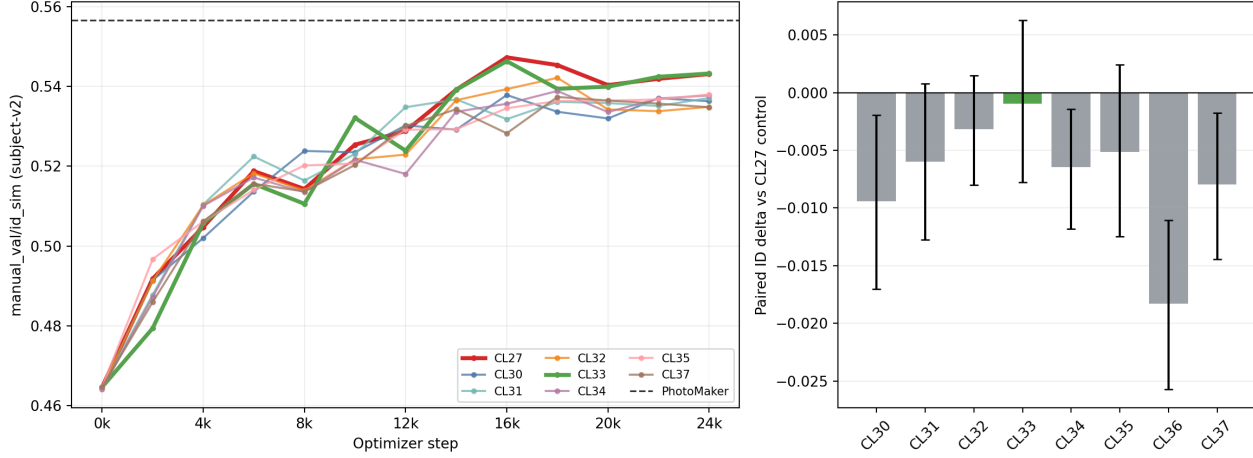


Figure 1. Left: complete subject-v2 ID trajectories for CL27 and the full 24k arms; PhotoMaker is the dashed line. Right: each arm’s selected-checkpoint paired difference against the correct CL27 control with 95% cell-bootstrap intervals. CL36 is compared with its CL27-16k source.

Arm	Selected ID @ step	Delta versus control	Wins	95% interval
CL27	0.547260 @16k	—	—	—
CL30	0.537826 @16k	-0.009435	38/96	[-0.017018, -0.001939]
CL31	0.537079 @24k	-0.006001	43/96	[-0.012779, +0.000756]
CL32	0.542138 @18k	-0.003182	45/96	[-0.008031, +0.001499]
CL33	0.546311 @16k	-0.000949	47/96	[-0.007764, +0.006280]
CL34	0.538839 @18k	-0.006482	47/96	[-0.011836, -0.001425]
CL35	0.537951 @24k	-0.005130	37/96	[-0.012482, +0.002413]
CL36	0.528958 @4k	-0.018302	32/96	[-0.025719, -0.011047]
CL37	0.537343 @18k	-0.007978	35/96	[-0.014435, -0.001741]
PhotoMaker	0.556580 @0	+0.009320 vs CL27	50/96	[-0.003438, +0.021757]

CL30, CL34, CL36, and CL37 are clearly worse. CL31 and CL35 have negative point estimates but intervals crossing zero; CL32 is neutral; CL33 is the closest result but supplies no evidence for promotion. Final 24k ID values for CL30–CL37 are, respectively, 0.536273, 0.537079, 0.534867, 0.543207, 0.537722, 0.537951, 0.528958 @4k, and 0.534766. [measured]

The trajectories repeat the CL27 early-stop lesson. CL27 and CL33 both peak at 16k and lose about 0.0042 and 0.0031, respectively, by 24k. CL32, CL34, and CL37 peak at 18k and also decline. Future 24k jobs should retain validation through 24k for comparability but treat 16k and 18k as required promotion gates rather than assuming the final checkpoint is best. [measured] [decision]

2.2 Hard-case and identity slices

Slice	PhotoMaker	CL27	CL30	CL31	CL32	CL33	CL34	CL35	CL36	CL37
Skiing	0.4640	0.4337	0.4145	0.4158	0.4075	0.4620	0.4156	0.4175	0.3928	0.4118
Crying	0.6000	0.5855	0.5561	0.5695	0.5680	0.5710	0.5611	0.5740	0.5658	0.5689
Jumping	0.4173	0.3946	0.3871	0.4099	0.4105	0.4046	0.3978	0.3933	0.3784	0.3930
Dancing	0.4487	0.4422	0.4324	0.4189	0.4321	0.4496	0.4227	0.4619	0.4342	0.4219
Marion, all prompts	0.5029	0.4935	0.4716	0.4885	0.4747	0.4819	0.4737	0.4599	0.4593	0.4789

CL33's +0.028303 Skiing gain over CL27 is real in the metric, but is not accompanied by Crying or Marion gains: those fall by 0.014535 and 0.011558. CL32's Jumping result and CL35's Dancing result are useful cell-family signals, not base promotions. No new arm closes CL27's Marion gap. [measured]

2.3 Face ownership and quality

Arm	Mask IoU mean	IoU median	IoU p10	Mean faces	Multi-face cells	TOPIQ-Face mean / p10
PhotoMaker	0.8652	0.8825	0.7772	1.135	12	0.7532 / 0.5918
CL27	0.9211	0.9358	0.8858	1.125	11	0.7142 / 0.5882
CL30	0.9259	0.9346	0.8793	1.135	13	0.7077 / 0.5874
CL31	0.9247	0.9326	0.8894	1.135	11	unavailable
CL32	0.9257	0.9377	0.8840	1.115	11	0.7131 / 0.5817
CL33	0.9205	0.9290	0.8739	1.115	10	0.7140 / 0.5797
CL34	0.9252	0.9335	0.8821	1.115	10	0.7157 / 0.5891
CL35	0.9234	0.9354	0.8848	1.104	9	0.7146 / 0.5844
CL36	0.9197	0.9255	0.8803	1.115	11	0.7125 / 0.5901
CL37	0.9248	0.9310	0.8807	1.094	8	0.7159 / 0.5856

All selected BA panels have zero cells below 0.3 mask IoU. There is no broad face/body relocation collapse. BA continues to improve intended-box ownership relative to PhotoMaker, while PhotoMaker retains the higher average face quality and identity. Eddie/Dancing remains a useful low-IoU stress cell: PhotoMaker scores 0.341, CL27 0.448, and the new BA arms about 0.418-0.455. [measured]

CL31's face-quality series is absent from Comet, so its values are reported as unavailable rather than inferred from other runs. [measured] [caveat]

3. Visual inspection: critical differentiators

3.1 Skiing exposes a CL33 shortcut

	PM0 · 0k	CL27 · 16k	CL30 · 16k	CL31 · 24k	CL32 · 18k
eddie idx 2	 ID 0.303 · IoU 0.763	 ID 0.461 · IoU 0.887	 ID 0.436 · IoU 0.894	 ID 0.404 · IoU 0.898	 ID 0.416 · IoU 0.890
elon idx 14	 ID 0.444 · IoU 0.933	 ID 0.385 · IoU 0.959	 ID 0.368 · IoU 0.969	 ID 0.367 · IoU 0.951	 ID 0.375 · IoU 0.978
jennie idx 26	 ID 0.663 · IoU 0.951	 ID 0.644 · IoU 0.950	 ID 0.569 · IoU 0.958	 ID 0.633 · IoU 0.946	 ID 0.564 · IoU 0.974
jensen idx 38	 ID 0.486 · IoU 0.772	 ID 0.493 · IoU 0.902	 ID 0.480 · IoU 0.981	 ID 0.480 · IoU 0.929	 ID 0.526 · IoU 0.913
jisoo idx 50	 ID 0.601 · IoU 0.929	 ID 0.493 · IoU 0.915	 ID 0.511 · IoU 0.939	 ID 0.472 · IoU 0.923	 ID 0.457 · IoU 0.945
keanu idx 62	 ID 0.402 · IoU 0.756	 ID 0.419 · IoU 0.850	 ID 0.414 · IoU 0.863	 ID 0.317 · IoU 0.838	 ID 0.398 · IoU 0.823
lex idx 74	 ID 0.393 · IoU 0.828	 ID 0.316 · IoU 0.951	 ID 0.349 · IoU 0.948	 ID 0.378 · IoU 0.962	 ID 0.330 · IoU 0.964
marion idx 86	 ID 0.420 · IoU 0.911	 ID 0.258 · IoU 0.909	 ID 0.190 · IoU 0.903	 ID 0.275 · IoU 0.917	 ID 0.194 · IoU 0.897

Figure 2. Skiing at each selected checkpoint: PhotoMaker, CL27, and CL30–CL32. Every cell is the face crop from the fixed ownership box; values are per-image subject-v2 ID and mask IoU.

	CL27 · 16k	CL33 · 16k	CL34 · 18k	CL35 · 24k	CL36 · 4k	CL37 · 18k
eddie idx 2	 ID 0.461 · IoU 0.887	 ID 0.505 · IoU 0.833	 ID 0.436 · IoU 0.863	 ID 0.418 · IoU 0.906	 ID 0.390 · IoU 0.891	 ID 0.452 · IoU 0.869
elon idx 14	 ID 0.385 · IoU 0.959	 ID 0.521 · IoU 0.984	 ID 0.379 · IoU 0.970	 ID 0.355 · IoU 0.935	 ID 0.299 · IoU 0.963	 ID 0.369 · IoU 0.983
jennie idx 26	 ID 0.644 · IoU 0.950	 ID 0.617 · IoU 0.970	 ID 0.576 · IoU 0.965	 ID 0.595 · IoU 0.983	 ID 0.583 · IoU 0.958	 ID 0.612 · IoU 0.962
jensen idx 38	 ID 0.493 · IoU 0.902	 ID 0.506 · IoU 0.912	 ID 0.508 · IoU 0.920	 ID 0.497 · IoU 0.927	 ID 0.468 · IoU 0.903	 ID 0.492 · IoU 0.881
jisoo idx 50	 ID 0.493 · IoU 0.915	 ID 0.515 · IoU 0.903	 ID 0.491 · IoU 0.963	 ID 0.537 · IoU 0.933	 ID 0.484 · IoU 0.918	 ID 0.395 · IoU 0.971
keanu idx 62	 ID 0.419 · IoU 0.850	 ID 0.428 · IoU 0.881	 ID 0.398 · IoU 0.877	 ID 0.402 · IoU 0.872	 ID 0.377 · IoU 0.870	 ID 0.349 · IoU 0.882
lex idx 74	 ID 0.316 · IoU 0.951	 ID 0.375 · IoU 0.965	 ID 0.338 · IoU 0.959	 ID 0.296 · IoU 0.941	 ID 0.315 · IoU 0.954	 ID 0.372 · IoU 0.985
marion idx 86	 ID 0.258 · IoU 0.909	 ID 0.228 · IoU 0.888	 ID 0.198 · IoU 0.896	 ID 0.240 · IoU 0.902	 ID 0.227 · IoU 0.909	 ID 0.253 · IoU 0.890

Figure 3. Skiing at the selected checkpoints for CL27 and CL33–CL37.

Run	Pass	Minor	Fail
PhotoMaker	8	0	0
CL27-16k	6	1	1
CL33-16k	6	0	2

PhotoMaker keeps every large-goggle layer visibly above a readable face and retains the male identities’ ordinary glasses. CL27 has localized nested- eyewear asymmetry for Eddie and a clear Marion failure. CL33 improves several large-goggle layers, but Elon’s ordinary glasses disappear and Marion still fails. [visual]

CL33 raises Eddie from 0.461 to 0.505, Elon from 0.385 to 0.521, Jisoo from 0.493 to 0.515, and Keanu from 0.419 to 0.428. Elon’s jump is not a valid topology win because it removes a required identity object. Jisoo and Keanu are cleaner examples of the visibility-balanced loss helping without obvious deletion. Marion remains unresolved in every arm. [measured] [visual]

CL34 and CL35 often keep readable large-goggle order, but their aggregate Skiing means stay below CL27. CL32 introduces localized rim/eye corruption in some rows. CL37 gives no consistent small-face or occlusion benefit. The metric mean alone would have promoted CL33’s Skiing behavior; the topology rubric correctly prevents that mistake. [visual]

3.2 Crying is stable, not the remaining bottleneck



Figure 4. Crying for CL27 and CL33–CL37; hands and face boundaries remain readable.

The previous hand/face fusion failure does not recur. All eight reviewed Crying rows pass for PhotoMaker, CL27, and CL33; the other selected panels are also visually coherent. Differences are mainly identity strength rather than topology. For example, CL27 is stronger than CL33 on Jennie (0.624 versus 0.622) and Jensen (0.688 versus 0.678), consistent with the lower CL33 Crying mean. Further architecture work should not trade this recovered behavior for Skiing gains. [measured] [visual]

3.3 Small/action faces: isolated gains do not generalize



Figure 5. Base-decision cells across PhotoMaker, CL27, CL32, CL33, and CL36: Jisoo/Keanu Skiing, Jennie/Jensen Crying, Eddie Dancing, and Marion Night-ride. Each column shows the full image beside the owned-face crop.

CL32 improves several Jumping cells and reaches the best new Jumping mean, but does not improve aggregate ID or Skiing. CL36 demonstrates the risk of direct identity continuation without a verified gradient path: by 4k it has lost 0.018302 versus its exact CL27 source and visibly weakens several critical faces. [measured] [visual]

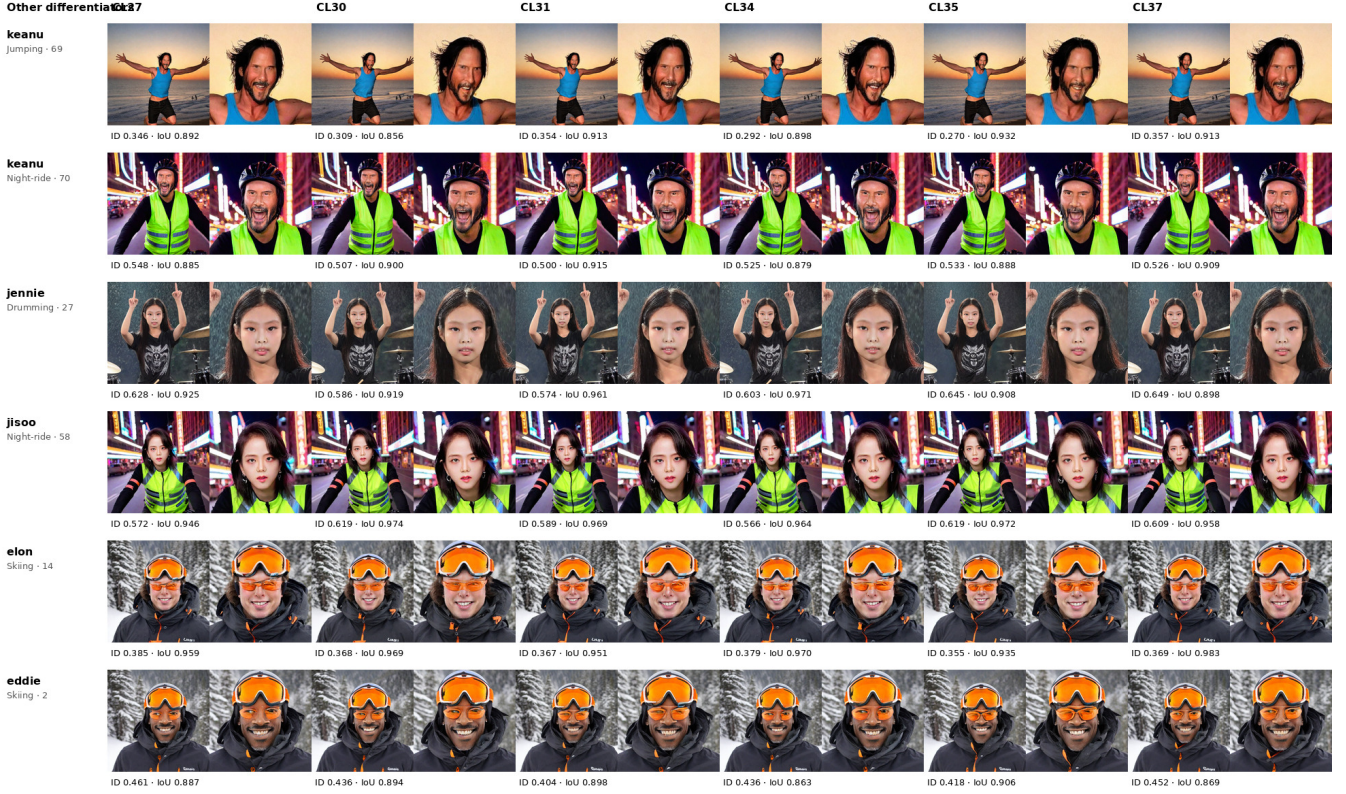


Figure 6. Additional differentiators across CL27, CL30, CL31, CL34, CL35, and CL37: Keanu Jumping/Night-ride, Jennie Drumming, Jisoo Night-ride, and Eddie/Elon Skiing.

CL37’s Keanu/Jumping cell rises from CL27 0.346 to 0.357, yet the whole Jumping mean falls slightly (0.3930 versus 0.3946) and aggregate ID is clearly worse. CL35’s Dancing mean is high, with strong Elon/Jisoo examples, but the inactive patch reward means the visual result cannot be attributed to DINO supervision. These are useful leads for eligibility and sampling design, not successful complete interventions. [measured] [visual]

4. What worked, what did not, and what was not actually tested

Arm	Was the new mechanism active?	Evidence and conclusion
CL30	yes	Positive pull is sampled at about 0.124; its loss is nonzero in 449/460 logged windows after 1k. It is clearly worse than CL27. Positive-only low-band attraction at this weight/cadence is rejected. [measured]
CL31	yes	Ownership loss is sampled at about 0.247 and nonzero in 449/461 windows. Aggregate effect is not established; some goggle order looks better, but there is no promotion. [measured] [visual]

Arm	Was the new mechanism active?	Evidence and conclusion
CL32	yes	Contact partition is applied at about 0.220. It produces the best new Jumping mean but lower Skiing and neutral aggregate ID. Contact repartition alone is insufficient. [measured]
CL33	yes	Visibility partition is applied at about 0.254. It preserves aggregate ID and improves mean Skiing, but sometimes optimizes by deleting ordinary glasses and lowers Crying/Marion. This is the only reusable objective idea, after anti-deletion repair. [measured] [visual]
CL34	technically, but barely moves	Learned schedule values change only 0.00085–0.00140 from initialization. The run is clearly worse and lacks exact step-zero parity. Do not reuse it as a base. [measured]
CL35	no, after startup	Observed gate mass is about 0.111 against a 0.55 floor, so patch-ID loss and applied fraction are zero after 1k. The complete recipe is negative; the intended DINO reward remains untested at sustained activity. [measured] [code]
CL36	mostly no	Applied fraction is about 0.0147; identity loss is nonzero in only 4/60 post-1k windows and BA auxiliary gradient ratio is always zero. Reject this continuation recipe, not ArcFace as a class. [measured] [code]
CL37	yes, but sparse	Teacher sampling is about 0.124, while eligibility averages only 0.0158. The signal is too sparse; one Keanu gain does not generalize. [measured] [visual]

The useful general lesson is **activity must be a promotion gate, not merely a logged diagnostic**. A loss can be configured, instantiated, and nonzero in a smoke while contributing nothing for almost all scientific training. Future identity or teacher objectives should fail closed before a full run if their conditional eligibility, nonzero-loss cadence, and BA-owned gradient ratio are outside a declared range. [decision]

5. Base configuration going forward

5.1 Decision: retain CL27, with a precise checkpoint contract

CL27 still makes sense as the base. The base should now be named precisely:

1. CL27 frequency-surface SA-only branched-attention architecture;
2. immutable source checkpoint CL27 r3 @16k, not the 24k endpoint;
3. target queries consuming reference K/V, with `pose_adapt_ratio=0`, `ca_mixing_for_face=false`, and unchanged reference weight;
4. optimized training pipeline and the fixed subject-v2 96-image validation contract;
5. explicit 16k and 18k selection gates even when training continues to 24k.

This is an operationally tightened CL27 base, not a new scientific configuration. It preserves the best established BA score and hard-case behavior while preventing a weaker 24k checkpoint from silently becoming the control. [measured] [decision]

5.2 Why no modified configuration should replace it

CL33 is the only plausible alternative, but it fails the promotion test for three independent reasons: aggregate ID improvement is not established; Crying and Marion regress; and its strongest Skiing gain includes ordinary-glasses deletion. Its 24k score also declines. Using CL33 as the default base would make later improvements harder to interpret because an anti-occlusion objective and its shortcut would already be entangled in every comparison. [measured] [visual] [decision]

CL31 and CL32 are mechanistically active but aggregate-neutral. CL30, CL34, and CL37 are clearly negative. CL35 and CL36 are confounded by effectively inactive intended rewards. None supplies evidence for a safer or higher-ID complete base. [measured]

The recommended arrangement is therefore:

- **primary control/base:** CL27-16k;
- **specialized secondary control:** CL33-16k only for a single-delta visibility-loss successor;
- **do not stack:** CL30/31/32/34/35/36/37 mechanisms into the base before a corrected isolated arm passes both metric and visual gates.

6. Highest-value next work

Priority 1 — visibility-balanced v2 with anti-deletion ownership

Start cold from the CL27 configuration and change only the reconstruction weighting. Preserve CL33's visible-face/top-object/contact partition, but add a small object-presence/ownership term that penalizes erasing identity-defining ordinary eyewear when it is visible in the target. The term should supervise the top-object region and ordinary-eyewear region separately so the model cannot improve the large-goggle face crop by deleting the smaller glasses. [hypothesis]

Promotion gates: step-zero exact parity; sustained partition activity; aggregate ID not below CL27 at both 16k and 18k; Skiing at least 7/8 passes; zero ordinary-glasses deletions; Crying and Marion no worse than CL27 within predeclared tolerances. The main risk is noisy ordinary-eyewear labeling.

Priority 2 — requalify patch identity before another 24k run

Do not rerun CL35 unchanged. First use a bounded diagnostic to set eligibility from the observed gate-mass distribution rather than the impossible 0.55 floor. Require nonzero reward in a declared fraction of eligible low-noise steps and a nonzero gradient ratio on BA-owned trainables. Only then run one CL27-based arm with the same DINO objective. [measured] [hypothesis]

Promotion gates: no fallback to ordinary three-reference CL31 behavior; conditional application and gradient telemetry in range; no aggregate ID loss at 4k/8k early gates; no Skiing object deletion. This tests the intended loss, which CL35 did not.

Priority 3 — make small-face teaching eligible before increasing weight

CL37's ~1.58% eligible fraction is too sparse. A successor should change only the eligibility/sampling design—for example, stratify batches by measured face area or compute the teacher on all sampled small-face records—while retaining the same stop-gradient teacher and weight. Do not increase weight until the current signal is observed often enough to evaluate. [measured] [hypothesis]

Promotion gates: declared eligible fraction reached; Jumping and Dancing both improve without lowering whole-panel ID; face/body ownership and detected face count do not regress.

ArcFace should remain behind these priorities. Before any scientific rerun, the exact auxiliary gradient path must be demonstrated on BA-owned parameters; CL36's zero gradient ratio makes a new 24k launch unjustified. [decision]

7. What is established and what is not

Established: no completed CL30–CL37 arm improves aggregate ID over its correct CL27 control; CL33 is the closest and raises Skiing mean; PhotoMaker still has the higher mean ID and cleanest Skiing topology; BA panels retain better intended-box ownership; Crying topology remains repaired; Marion and small/action faces remain inconsistent. [measured] [visual]

Not established: CL33 is equivalent across training seeds; DINO patch identity or ArcFace supervision is intrinsically ineffective; CL37's teacher would fail under adequate eligibility; the one-seed visual rubric predicts multi-seed user preference. [limitation]

Decision confidence: high that CL27-16k should remain the primary base; moderate that visibility-balanced v2 is the best next scientific arm; low on the value of direct identity losses until their gradient path is proven.

8. Reproduction

From `diffusion_template/`, the immutable report inputs are the final and peak Comet manifests in `analysis/assets/cl30_cl37_20260819/`. The selected/final per-image tables, joined slice data, paired comparisons, step-zero audit, telemetry summary, visual-review counts, figure builder, and file hashes are in the same asset directory.

```
python analysis/assets/cl30_cl37_20260819/build_analysis_assets.py
sha256sum -c analysis/assets/cl30_cl37_20260819/SHA256SUMS.txt
python tools/reports/publish_report.py \
    analysis/2026-08-19_cl30_cl37_completed_results_and_base_decision.md --upload
```

The bulk downloaded Comet images are reproducible cache and are intentionally not part of the repository artifact. The curated grids and exact tables are the durable visual and numeric evidence. [reproducibility]

Prior internal evidence

- `analysis/2026-08-17_cl27_cl29_vs_cl23_visual_results_and_next_experiments.md` defines the CL27 selection and hard-case rubric.
- `analysis/2026-08-16_training_pipeline_processor_lookup_fix.md` defines the optimized scientific pipeline used by these runs.
- `docs/handoffs/LATEST.md` records immutable retries, source seals, and launch history; this report supersedes its provisional CL30–CL37 live-state entry.